THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which sequence is in increasing depth? 1

(a) core, mantle, crust **(b)** mantle, crust, core

(c) inner core, crust, mantle **(d)** crust, mantle, outer core, inner core

2. Which of these is true for the troposphere? 1

(a) It contains most of the water vapour and clouds

(b) It has no weather

(c) It contains the ozone layer

(d) It is the hottest layer

3. Water pollution reduces: 1

(a) availability of safe water **(b)** temperature

(c) rains **(d)** salinity of the oceans

4. Light from the Sun takes about ___ to reach the Earth. 1

(a) 8 days **(b)** 8 minutes **(c)** 8 seconds **(d)** 8 hours

5. Volcanic eruptions bring to the surface the material from the: 1

(a) mantle (magma) **(b)** inner core **(c)** ozone **(d)** atmosphere

6. The 'ozone hole' is most noticeable over: 1

(a) Africa **(b)** the equator **(c)** India **(d)** Antarctica

7. Salt water can be changed to drinking water by: 1

(a) adding sugar **(b)** freezing only **(c)** boiling only **(d)** desalination

8. The Earth's magnetic field is produced by the: 1

(a) ozone layer **(b)** oceans

(c) Moon **(d)** motion of the liquid outer core

9. Earthquakes are caused by: 1

(a) rain **(b)** winds

(c) sudden release of energy in the crust **(d)** tides

10. Which action helps protect the ozone layer? 1

(a) avoiding products with CFCs **(b)** burning plastics

(c) using more aerosols **(d)** cutting trees

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why should CFC-free refrigerators be used? 2

12. Why do aeroplanes fly in the lower stratosphere? 2

13. Why is the Earth's fresh water a limited resource? 2

14. Why are forests called the 'lungs of the Earth'? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how Earth's magnetic field protects life. 3

16. Explain why Earth's atmosphere is thicker near the surface. 3

17. Explain why light from the Sun is necessary for the Earth's food chains and climate. 3

OR

Explain the use of the ozone layer and the role of Montreal Protocol. 3

18. Explain how the Earth's tilt and revolution lead to seasons and why they matter for life. 3

OR

Explain how a rise of 2 degrees Celsius in average temperature could affect agriculture and water. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss the distribution of water on Earth and the importance of managing fresh water with five methods of conservation. 5

OR

Explain sustainable use of the Earth's resources with at least five measures and the idea of renewable resources. 5

20. Explain how the four spheres interact in the formation of rain, soil and a forest ecosystem. 5

OR

Explain how the Earth's rotation, revolution and tilt produce the day, night, year and seasons in terms of conditions for life. 5

END OF QUESTION PAPER — Total: 40 Marks

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